Axiom-Zero: Technical Research Documentation

Fateh Abderrahim BOUKHALFA
Trustworthy AI Systems Engineering Research

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Abstract

Axiom-Zero addresses the critical challenge of algorithmic accountability in high-impact macro-governance. This research explores how modular AI architectures can support human decision-making by providing traceable, explainable, and auditable recommendations in complex socio-economic contexts.

Research Problem

Modern automated governance systems often operate as "black boxes." The core problem investigated herein is: **How can AI support high-impact governance decisions while ensuring the process remains interpretable and challengeable by human stakeholders?**

System Methodology & Design

Axiom-Zero is engineered around three primary design principles:

- **Modular Decoupling:** Isolating data ingestion, decision logic, and output validation to allow for independent auditing.
- **Traceability Layer:** Implementing a "logic-log" that records the weighted parameters leading to each recommendation.
- **Human-in-the-Loop (HITL) Integration:** Designing decision nodes that require human confirmation, framing the AI as a consultative partner.

Experimental Approach & Challenges

- **The Convergence Challenge:** During testing, the system faced instability when processing high-velocity macroeconomic data.
- **Resolution:** The methodology shifted toward **Constraints-Based Modeling**, ensuring that the AI operates within strict boundary conditions defined by human policy.

Engineering Reflection

This project investigates the trade-off between AI performance and transparency. Findings suggest that explainability is a system requirement, not an optional byproduct. True trustworthy AI is achieved through architectural accountability.

Future Scope

Axiom-Zero serves as a sandbox for:

- Developing auditable decision trails.
- Establishing stability benchmarks for AI-assisted governance.
- Protocols for human-AI interaction in high-impact environments.

Project Repository: https://github.com/YourGitHubProfile/Axiom-Zero